

Template for your Machine Learning (ML) final paper

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Abstract—These instructions give you guidelines for preparing final papers for Machine Learning (ML) course of the UNESP Electrical Engineering Graduate Program, based on IEEE Transactions and Journals. Use this document as a template if you are using Latex. Paper titles should be written in uppercase and lowercase letters, not all uppercase. Full names of authors are preferred in the author field, but are not required. Put a space between authors' initials. The ML student is the first author of the paper. You can include other people as co-authors. Typically, you should invite your advisor to be one of the co-authors. The ML professor is also an author. The abstract must be a concise yet comprehensive reflection of what is in your article. In particular, the abstract must be self-contained, without abbreviations, footnotes, or references. It should be a microcosm of the full article. The abstract must be between 150–250 words. Be sure that you adhere to these limits. The abstract must be written as one paragraph, and should not contain displayed mathematical equations or tabular material. The abstract should include three or four different keywords or phrases, as this will help readers to find it. It is important to avoid over-repetition of such phrases as this can result in a page being rejected by search engines. Ensure that your abstract reads well and is grammatically correct.

Keywords—Robotics, Machine Learning, Artificial Neural Networks (ANNs).

I. INTRODUCTION

In this section you should introduce all important aspects of your work. You should include citations using latex references, like in [1]. You can also insert figures (like in figure 1) or tables (like in table I). You can insert as many sections and subsections you want.

The introduction section is expected to present details about the domain you are working on, your goals and some other general information. Some important points are:

- 1) Your **motivation**: why is this topic important? Why this problem is relevant?
- 2) A clear and understandable **explanation about your problem**, the context where it is inserted;



Figure 1. This is the place where I live.

Table I
EXAMPLE OF LATEX TABLE.

Application	Functionalities						Place		
	Sensors	Fixing	Camera	Manipulator	Dispenser	Sprinkler	Spray	Indoor	Outdoor
Mapping	✓		✓					✓	✓
Transport			✓		✓	✓			✓

- 3) A review of the **main techniques available in the literature** that are/were employed by others in this domain (your research should be focused on papers of the most relevant conferences and journals), and their most important results;
- 4) Your **goals**, a brief description about **your approach** to solve the problem and an explanation about how it is inserted in the literature (highlighting the main differences between your approach and others);
- 5) A quick description about the contents of your **sections** in this paper.

II. LITERATURE REVIEW

This section should contain a strong theoretical review about all the **machine learning technique(s)** that are relevant to this work, particularly focusing on those you intent to apply, with all relevant aspects of the theory, formulas, and so on. An example of equation is shown in equation 1.

$$y = x^{2+i} + x^2 + 1 \quad (1)$$

Don't forget to describe all your variables after the equations.

III. PROPOSED APPROACH

The **proposed approach** is the section where you should detail how you will apply your ML technique in your domain. Start describing the big picture, and only after this general overview describe the approach details. Please insert your own block diagrams and figures to make sure that the reader is able to understand all aspects of your approach.

IV. MATERIALS AND METHODS

The **materials and methods** section should contain all details about your experiments (yes... experiments are mandatory!). Do not forget to discuss your input data and your algorithm's setup. Reference your dataset if you are using a public one. Keep in mind that the reader of your paper should be able to fully replicate your experiment with the information you provide in this section. Please, be sure you inserted all values, constants, preprocessing methods and all information required to allow the full replication of your results. The implementation code of previously known algorithms should not be inserted in the paper (the algorithm should always be addressed from a theoretical point of view). You are expected to insert some algorithm (in the format shown in algorithm 1) if you have some important improvement or adaptation in a previously known algorithm.

V. RESULTS AND DISCUSSION

The **results** section presents all important graphs, images, tables and their (quantitative and qualitative) analysis. Do

Algorithm 1: This is my new algorithm.

```
Result: Write here the result
initialization;
while While condition do
    instructions;
    if condition then
        instructions1;
        instructions2;
    else
        instructions3;
    end
end
```

not forget to insert all the curves of the convergence of your algorithms, as well as the results. Do not forget to compare your results with those obtained with other approaches (discussed in item 3 of section I) and to highlight the main differences between both results. You are expected to employ some metrics to evaluate your algorithms and compare them to the values of the evaluation of other algorithms. Nice graphics are always important to enhance your work.

VI. CONCLUSIONS

In this section you should insert your conclusions. Refer to your goals and discuss about your successful rate with your approach.

ACKNOWLEDGEMENTS

Here you should insert your acknowledges.

REFERENCES

- [1] D. Adams, *The Hitchhiker's Guide to the Galaxy*. San Val, 1995.